

# S. PADMANABHAN

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## SUMMARY

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Junior AI-engineer specializing in Computer Vision and Applied Artificial Intelligence with hands-on experience in building production-oriented Deep learning systems. Strong background in dataset engineering, model validation, deep learning workflows, and AI application deployment. Proficient in Python, Machine Learning, YOLO, OpenCV, FastAPI, and Generative AI technologies, with a proven track record of contributing to high-impact projects involving image classification, object detection, OCR, and intelligent Gen-AI application. Seeking opportunities to develop scalable AI products and advance state-of-the-art machine learning solutions.

## EXPERIENCE

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### Junior AI Engineer

Way.com

Feb 2025 – Present

Thiruvananthapuram, Kerala

- Built and maintained a large-scale vehicle image dataset of 36,000+ annotated images, achieving 95%+ annotation quality for YOLO model training readiness.
- Led end-to-end computer vision pipeline for vehicle damage detection, coordinating dataset preparation, model validation, testing, and deployment activities.
- Conducted extensive literature reviews of state-of-the-art Computer Vision research papers and benchmarked multiple architectures to identify optimal solutions.
- Collaborated closely with AI engineers and developers to validate AI models and support successful deployment of AI-powered web applications.

## PROJECTS

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### IntelliDocs – RAG Document Q&A | Python, LangChain, FAISS, Groq API, Streamlit

- Built an end-to-end RAG pipeline enabling natural language querying over any PDF document.
- Implemented semantic search using FAISS vector store and HuggingFace embeddings (all-MiniLM-L6-v2).
- Integrated Groq API (LLaMA 3.3 70B) for fast, context-aware answer generation
- Deployed as a live web app on Streamlit Cloud
- Live Demo:[<https://ragdocumentq-intellidoc.streamlit.app/>]

### Vehicle Damage Detection System | Python, YOLOv8, OpenCV, EasyOCR, PyTorch

- Fine-tuned YOLOv8 on a custom dataset of 10,000+ vehicle images for multi-task detection and vehicle position classification.
- Integrated EasyOCR for number plate extraction and recognition with 91% accuracy.
- Prepared and validated large-scale annotation datasets containing 24,000+ images for model training and evaluation.
- Supported the complete model lifecycle including data collection, annotation, model training, validation, testing, and deployment.

## TECHNICAL SKILLS

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**Languages:** Python, SQL (Fundamentals)

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, OpenCV, Scikit-learn, TensorFlow, Keras

**Libraries:** Pandas, NumPy, Matplotlib

**LLMs & GenAI:** OpenAI APIs, RAG, Prompt Engineering, LLM Integration, FastAPI, LangChain, FAISS, Streamlit

**Tools:** Git, GitHub, Jupyter Notebook, VS Code, Image Annotation

## EDUCATION

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### PG Diploma in Data Science and AI

Edure Learning Pvt. Ltd.

Dec 2023 – Dec 2024

Thiruvananthapuram, Kerala

### B.Com Commerce with Computer Application

IHRD College of Applied Science

Aug 2020 – Apr 2023

Kollam, Kerala